

Traffic Noise Map of Quito DM-Ecuador

Traffic Noise Mapping & Public Health Assessment
- The Case Study of Quito, Ecuador — Bridging a
10-Year Regulatory Gap

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Secretaría de
Ambiente



Quito
Alcaldía Metropolitana



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A Decade-Long Institutional Delay & Regulatory Paradigm Shift

Historical Context: First Official Traffic Noise Map of Quito (2.6 million inhabitants across 32 urban parishes).



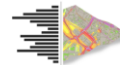
The Regulatory Gap: Local environmental ordinances required this traffic map over 10 years ago; execution was deferred due to technical and budgetary constraints.



Legacy Control vs. Modern Mapping:

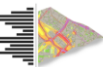
Legacy Framework (TULSMA Annex 5): Mandates strategic noise mapping for urban areas >150,000 residents.

Strategic Framework (Lambda / WHO 2018): Long-term spatial exposure modeling (Lden, Lnight) and population health burden.



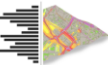
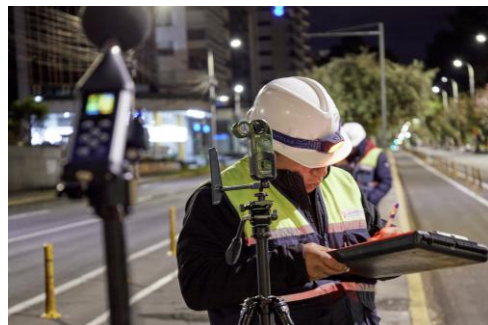
Physical Acoustics at 2,850 m a.s.l.: Topography & Microclimates

- Extreme Terrain: Narrow longitudinal valley (40 km N-S, 5 km E-W) bounded by mountains and deep ravines.
- Barometric & Atmospheric Influences: 2,850 m a.s.l. (~73 kPa barometric pressure) alters sound attenuation coefficients in air (α).
- Heavy Fleet Performance under Slopes: Road inclines exceeding 12% force heavy diesel transit into maximum torque, shifting spectral power to low frequencies (63–250 Hz).
- Morphological Diversity: Reverberant masonry canyons in the UNESCO Historic Center vs. high-speed ring roads (Av. Simón Bolívar).



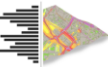
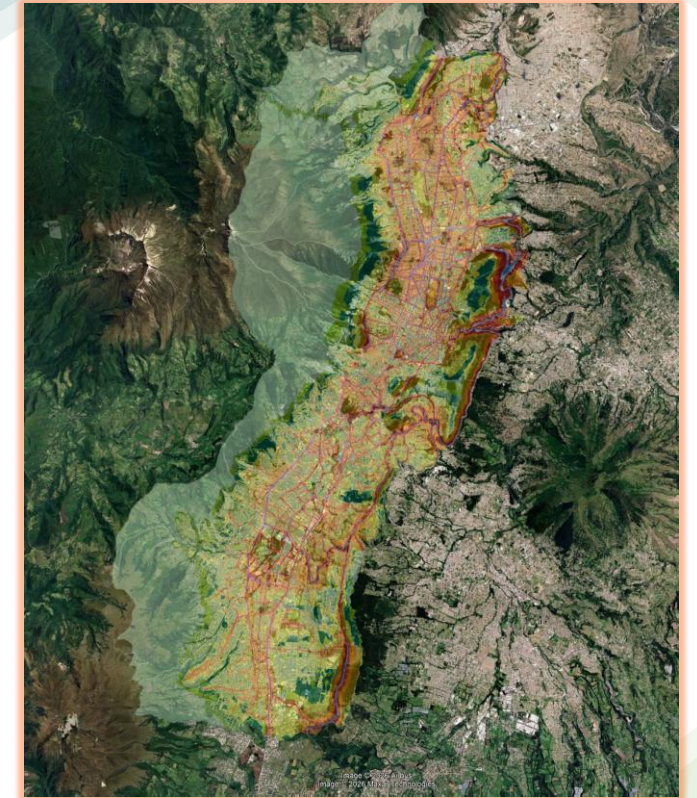
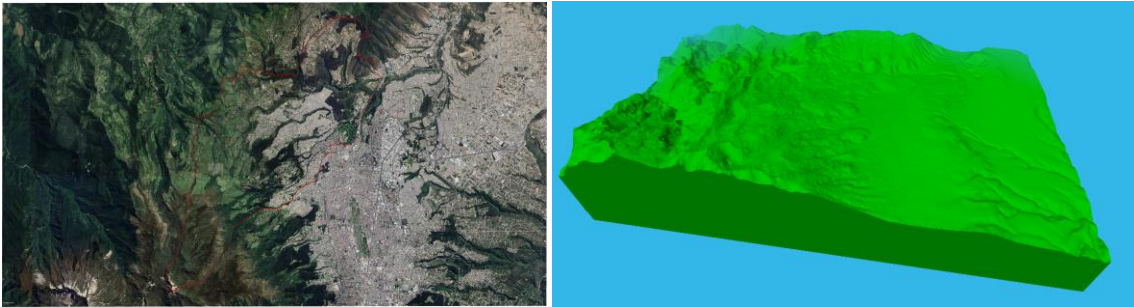
GIS-Driven Sampling Strategy & Metrological Protocol (160 Strategic Points)

- GIS-Based Site Selection (Non-Empirical): Spatial sampling optimization across 32 urban parishes integrating the Land Use and Management Plan (PUGS), road network hierarchy, and 3D building/receptor layers.
- Advanced Spatial Analytics: Application of Kernel Density Estimation (KDE) and Optimized Hotspot Analysis to ensure statistically significant spatial representativity.
- Instrumentation: Class 1 Sound Level Meters (Svantek SVAN 977/971) with active weather protection and continuous 1/3 octave logging.
- Quality Assurance (ISO/IEC 17025): Pre/post field acoustic calibrations using Class 1 calibrators and expanded Uncertainty Budget: Evaluated at $U = \pm 3.2$ dB ($k=2$, 95% confidence level).



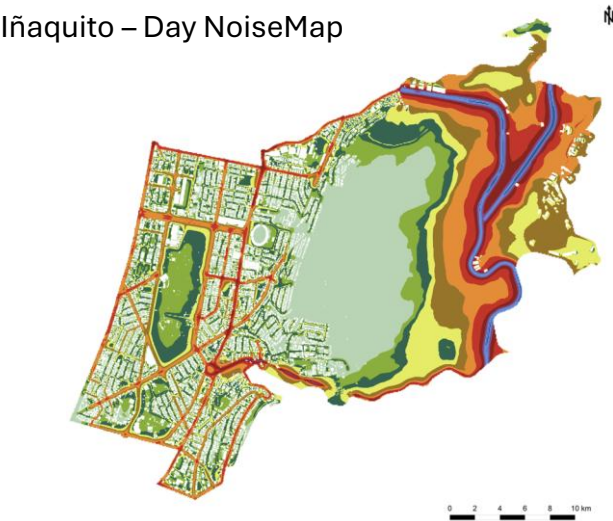
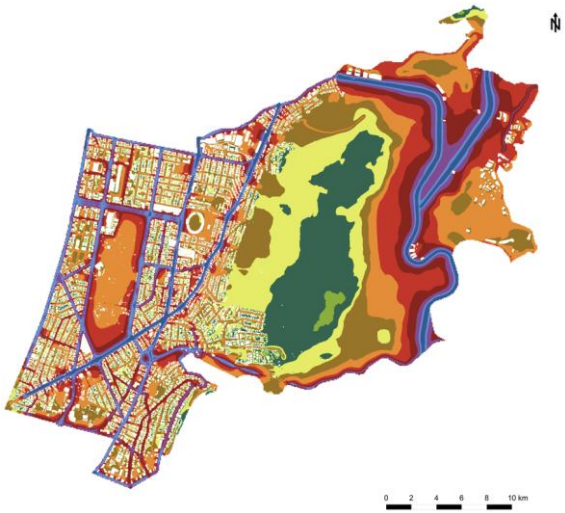
CAD / GIS Modeling Pipeline & Model Validation

- 3D Spatial Dataset: Digital Terrain Model (DTM), 310.101 receiver building vectors with individual height data, and road network attributes.
- Propagation Algorithm: RLS-90 (German Road Noise Algorithm).
- Emission Parameters: Traffic volume, heavy/light mix, vehicle speeds, gradient corrections (>12%), and pavement absorption (G).
- Model Validation Target: Iterative calibration against 160 points achieving RMS error $\leq \pm 3$ dB(A).



Spatial Inmission Trends Across 5 Zonal Administrations

Zonal Admin	Max Day LAeq,d [dBA]	Critical Corridor Location	Max Night LAeq,n [dBA]	Night Drop (Delta Leq)
La Delicia	88.7	Av. Simón Bolívar / Carcelén	74.0	14.7 dB
Manuela Sáenz	88.0	Playón de La Marín / Av. Pichincha	85.8	2.2 dB (Av. Simón Bolívar)
Eugenio Espejo	87.7	Av. Río Coca / Jipijapa	69.3	18.4 dB
Quitumbe	87.8	Av. Maldonado / Turubamba	81.1	6.7 dB
Eloy Alfaro	84.9	Av. Tnte. Hugo Ortiz	65.6	19.3 dB



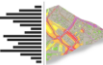
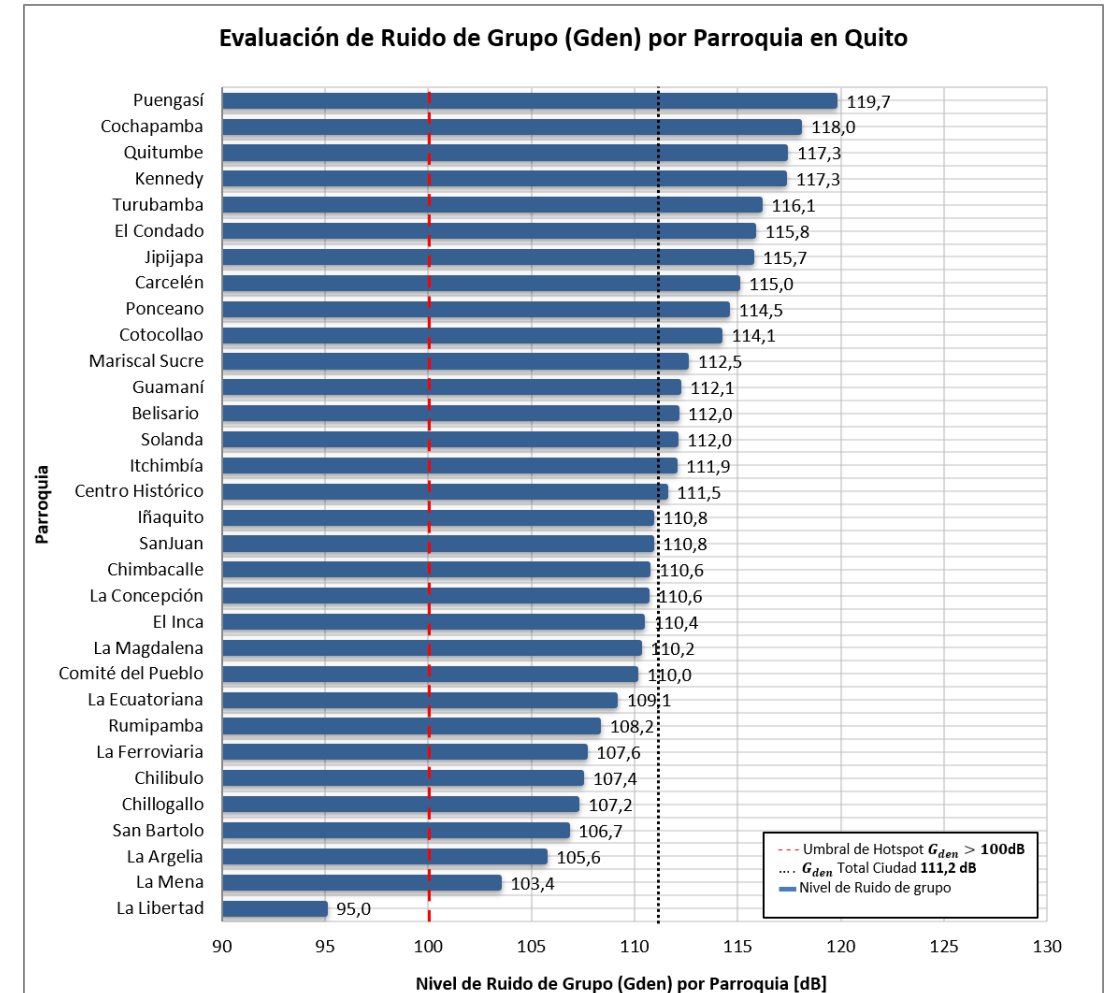
Prioritizing Hotspots: Group Noise Level (Gden & Gden,norm)

- Group Noise Level Formulation:
Evaluates cumulative energy exposure across 310,101 residential units.

$$G_{den} = 10 \log_{10} \left(\sum_i n_i \cdot 10^{\frac{L_{den,i}}{10}} \right) [\text{dB}]$$

- Hotspot Ranking Results:

Parish	Lden	Dwellings	Gden	Gden,norm
Puengasí	78,7	12.598	119.7	64.8
Cochapamba	78,2	9.536	118.0	63.1
Quitumbe	76,6	11.651	117.3	62.3



Quantifying Public Health Impact: %HA, %HSD & Sensitive Receivers

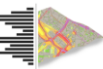
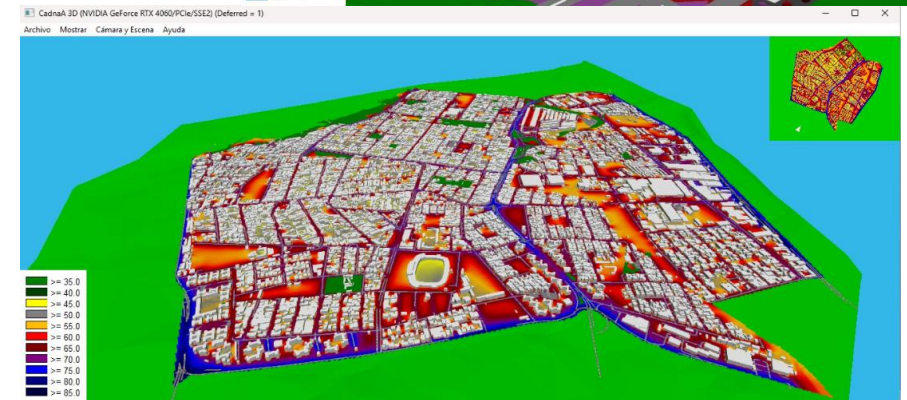
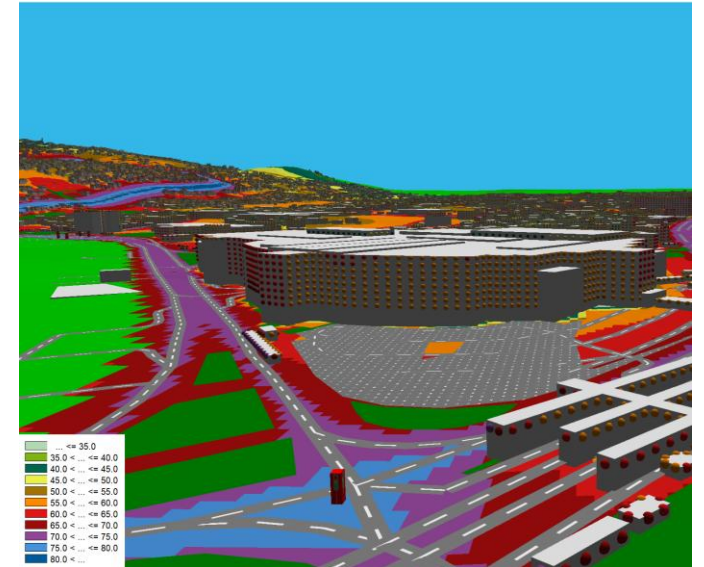
WHO 2018 Health Thresholds:

- Road Traffic Lden > 53 dB (>10% High Annoyance - %HA)
- Road Traffic Lnight > 45 dB (>3% High Sleep Disturbance - %HSD)

Exceedance Rate: Over 78% of monitoring points along arterial roads exceed the WHO 53 dB Lden threshold.

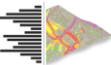
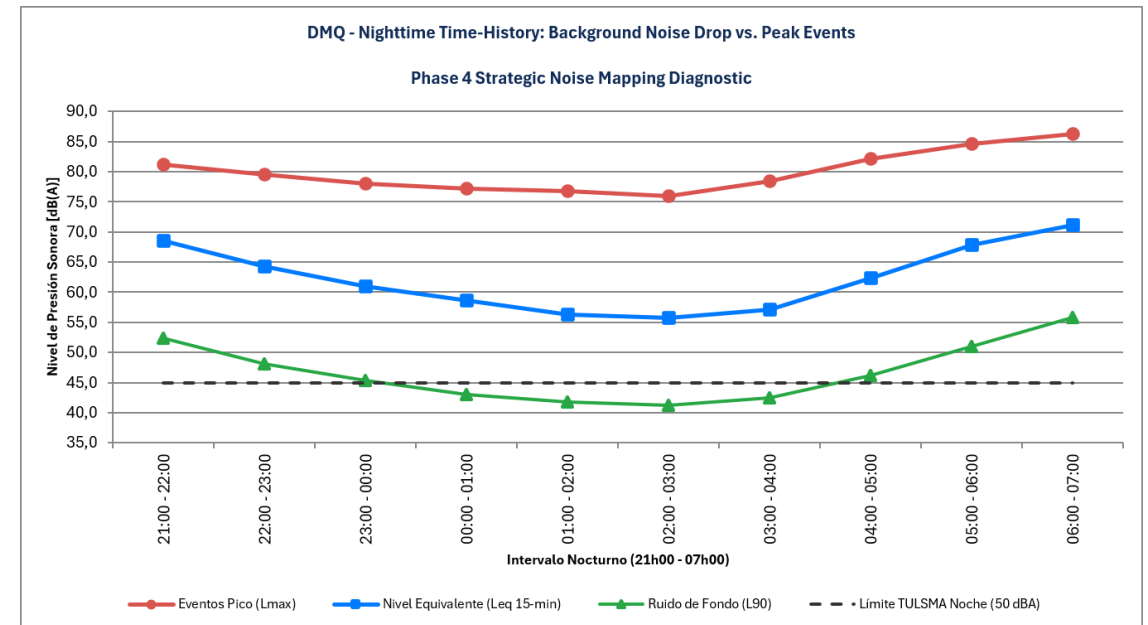
Land-Use Incompatibility: High levels (LAeq,d > 70 dB(A)) registered immediately adjacent to sensitive receptors

(e.g., Louis Víctor de Le Broglie School in Quitumbe, Montúfar College in Chimbacalle, SOLCA Hospital in San Isidro del Inca).



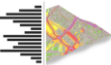
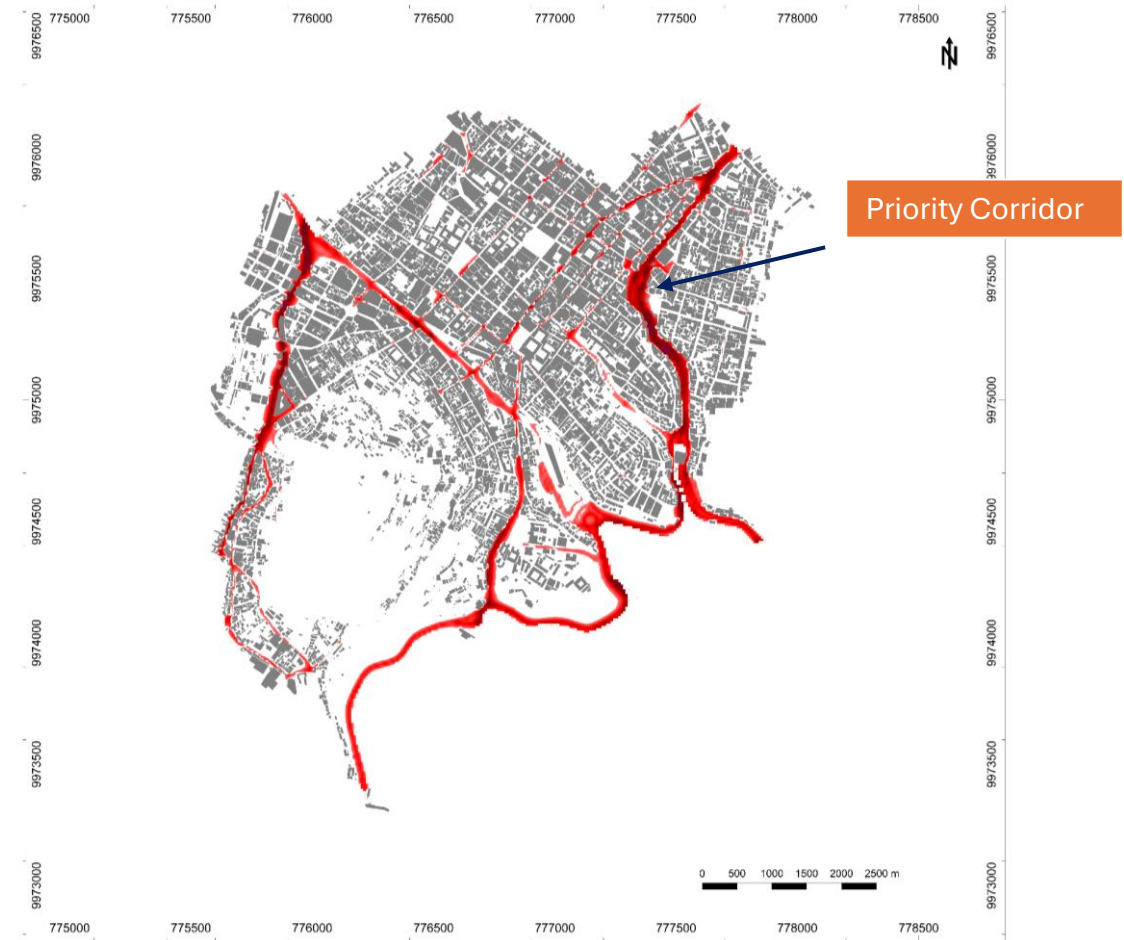
Critical Analytical Takeaways

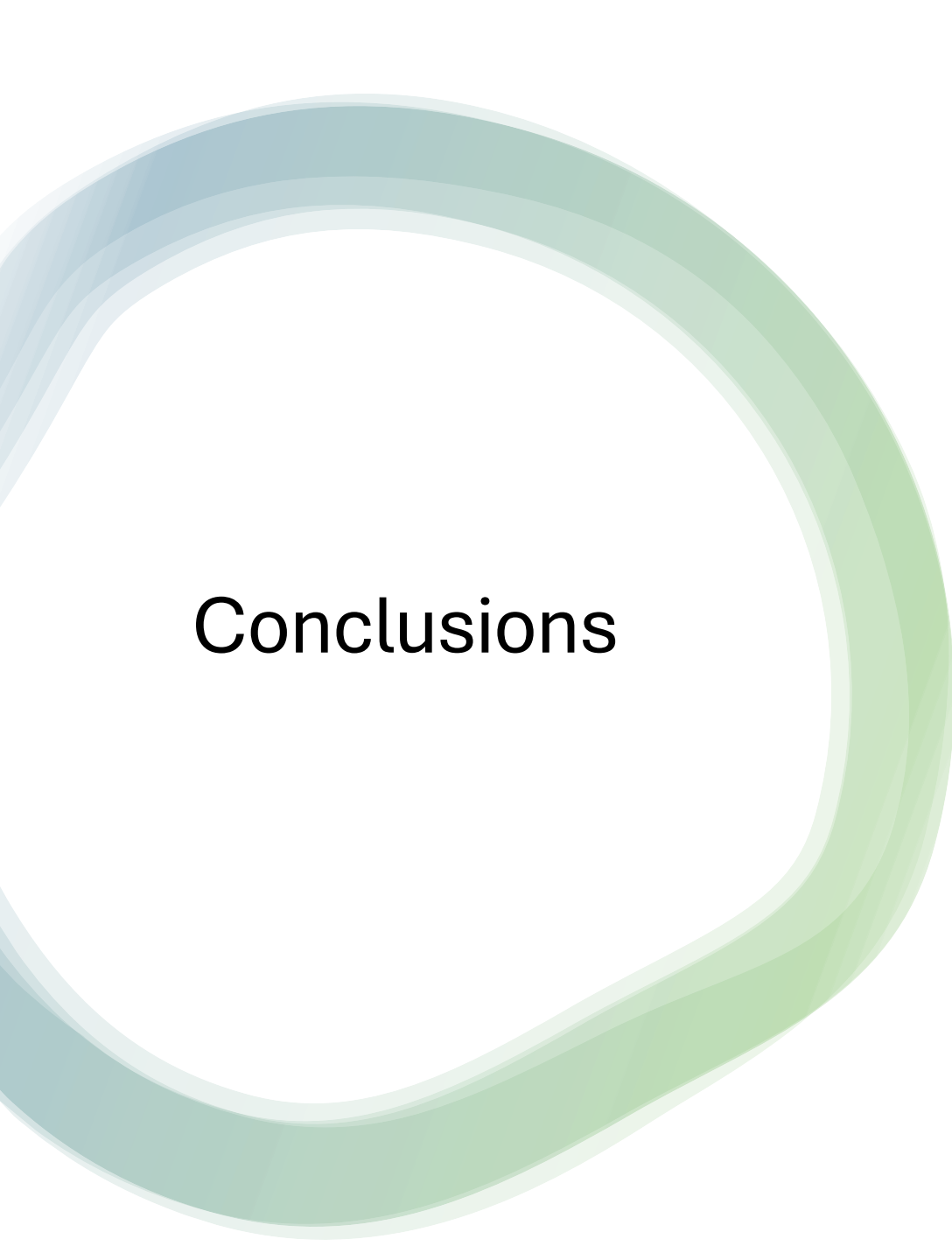
1. Infrastructure Dominance: Heavy traffic corridors dictate acoustic climate regardless of official land-use zoning. Secondary streets behind first-row buildings gain up to 30 dB structural shielding.
2. Geographical Stability: In >70% of parishes, the loudest daytime location remains the loudest at night due to continuous heavy freight movement.
3. Signal-to-Noise Ratio (SNR) Inversion: Sharp drops in nighttime background noise (30–35 dB) on secondary roads increase the acoustic prominence (Delta L) of isolated heavy vehicle pass-by, escalating sleep awakening risks (%HSD).



Evidence-Based Urban Mitigation Strategy for Quito

- Low-Emission & Low-Noise Zones (ZBE): Restricting heavy diesel freight access in high-density residential and historic centers during sleeping hours (21:00–06:00).
- Acoustic Pavements: Applying porous asphalt / SMA along high-speed corridors (Av. Simón Bolívar) to reduce tire-pavement interaction noise by 3–5 dB(A).
- Acoustic Building Codes: Updating municipal bylaws to mandate minimum composite facade sound insulation ($R'w + C_{tr} \geq 35$ dB) for new buildings within 50 m of primary arteries.
- Hybrid Continuous Monitoring Network: Expanding permanent acoustic stations to feed real-time calibration data into the city's GIS model.

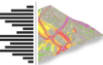




Conclusions

Key Technical Takeaways & Next Steps:

- Transition into open-source platforms like **NoiseModelling** and lot equipment
- Regulatory Resolution: Fulfills Quito's 10-year legal obligation under national environmental law.
- Prioritization Tool: Gden serves as the primary parameter to allocate public mitigation budgets.
- Mitigation Measures: Low-noise road resurfacing, Low Emission Zones (LEZ), traffic smoothing.
- Future Roadmap: Proposed expansion toward a continuous, hybrid acoustic monitoring network.



THANK YOU

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